

Exercise J01

Q1. The order and degree of the differential equation $= \left[4 + \left(\frac{dy}{dx} \right)^2 \right]^{2/3} = \frac{dy}{dx^2}$ are-

- (A) 2, 2 (B) 3, 3 (C) 2, 3 (D) 3, 2

Ans. C

Sol. -

Q2. The differential equation of all parabolas having their axis of symmetry coinciding with x-axis -

- (A) $y \frac{d^2}{dx^2} + \left(\frac{dy}{dx} \right)^2 = 0$ (B) $x \frac{d^2}{dx^2} + \left(\frac{dy}{dx} \right)^2 = 0$ (C) $y \frac{d^2 y}{dx^2} + \frac{dy}{dx} = 0$

- (D) none of these

Ans. A

Sol. -

Q3. The differential equation of system of concentric circles with centre (1, 2) is -

- (A) $(x - 2) + (y - 2) \frac{dy}{dx} = 0$ (B) $(x - 1) + (y - 2) \frac{dy}{dx} = 0$ (C) $(x + 1) \frac{dy}{dx} + (y - 2) = 0$

- (D) $(x + 2) \frac{dy}{dx} + (y - 1) = 0$

Ans. B

Sol. -

Q4. The order of the differential equation whose general solution is given by $y = (c_1 + c_2) \cos(x + c_3) - c_4 e^{x + c_5}$, where c_1, c_2, c_3, c_4, c_5 are arbitrary constant is -

- (A) 5 (B) 4 (C) 3 (D) 2

Ans. C

Sol. -

Q5. The value of the constant 'm' and 'c' for which $y = mx + c$ is a solution of the differential equation

$$D^2 y - 3Dy - 4y = -4x.$$

- (A) is $m = -1; c = 3/4$ (B) is $m = 1; c = -3/4$ (C) no such real m, c

- (D) is $m = 1; c = 3/4$

Ans. B

Sol. -

- Q6. The order & the degree of the differential equation whose general solution is, $y = c(x - c)^2$, are respectively
 (A) 1, 1 (B) 1, 2 (C) 1, 3 (D) 2, 1
- Ans. C
- Sol. -
- Q7. The differential equation, which represents the family of plane curves $y = e^{cx}$, is-
 (A) $y' = cy$ (B) $xy' - \log y = 0$ (C) $x \log y = yy'$ (D) $y \log y = xy'$
- Ans. D
- Sol. -
- Q8. The differential equation for the family of curves $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant is-
 (A) $2(x^2 - y^2) y' = xy$ (B) $2(x^2 + y^2) y' = xy$ (C) $(x^2 - y^2) y' = 2xy$
 (D) $(x^2 + y^2) y' = 2xy$
- Ans. C
- Sol. -
- Q9. The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where $c > 0$, is a parameter, is of order and degree as follows -
 (A) order 1, degree 2 (B) order 1, degree 1 (C) order 1, degree 3
 (D) order 2, degree 2
- Ans. C
- Sol. -
- Q10. The differential equation whose solution is $Ax^2 + By^2 = 1$, where A & B are arbitrary constants is of -
 (A) first order and second degree (B) first order and first degree
 (C) second order and first degree (D) second order and second degree
- Ans. C
- Sol. -
- Q11. The differential equation of all circles passing through the origin and having their centres on the x-axis is-
 (A) $x^2 = y^2 + xy \frac{dy}{dx}$ (B) $x^2 = y^2 + 3xy \frac{dy}{dx}$ (C) $y^2 = x^2 + 2xy \frac{dy}{dx}$ (D) $y^2 = x^2 - 2xy \frac{dy}{dx}$

Ans. C

Sol. -

Q12. The differential equation of the family of circles with fixed radius 5 units and centre on the line $y = 2$ is -

(A) $(y - 2) y'^2 = 25 - (y - 2)^2$ (B) $(y - 2)^2 y'^2 = 25 - (y - 2)^2$

(C) $(x - 2)^2 y'^2 = 25 - (y - 2)^2$ (D) $(x - 2) y'^2 = 25 - (y - 2)^2$

Ans. B

Sol. -

Q13. The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$ where c_1 and c_2 are arbitrary constants, is -

(A) $y' = y^2$ (B) $y'' = y' y$ (C) $yy'' = y'$ (D) $yy'' = (y')^2$

Ans. D

Sol. -

Q14. The solution of $\frac{dy}{dx} = e^x (\sin x + \cos x)$ is-

(A) $y = e^x (\sin x - \cos x) + c$ (B) $y = e^x (\cos x - \sin x) + c$ (C) $y = e^x \sin x + c$

(D) $y = e^x \cos x + c$

Ans. C

Sol. -

Q15. The solution of $\frac{dy}{dx} = x \log x$ is-

(A) $y = x^2 \log x - \frac{x^2}{2} + c$ (B) $y = \frac{x^2}{2} \log x - x^2 + c$ (C) $y = \frac{1}{2} x^2 + \frac{1}{2} x^2 \log x + c$

(D) None of these

Ans. D

Sol. -

Q16. The solution of the differential equation $\frac{dy}{dx} = (1 + x)(1 + y^2)$ is-

(A) $y = \tan(x^2 + x + c)$ (B) $y = \tan(2x^2 + x + c)$ (C) $y = \tan(x^2 - x + c)$

(D) $y = \tan\left(\frac{x^2}{2} + x + c\right)$

Ans. D

Sol. -

Q17. The general solution of the equation $\frac{d^2y}{dx^2} = -\frac{1}{x^2}$ is-

- (A) $y = \log x + c_1 x + c_2$ (B) $y = -\log x + c_1 x + c_2$ (C) $y = -\frac{1}{x} + c_1 x + c_2$
 (D) None of these

Ans. A

Sol. -

Q18. The solution of $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$ is-

- (A) $\log \left[1 + \tan \left(\frac{x+y}{2} \right) \right] + c = 0$ (B) $\log \left[1 + \tan \left(\frac{x+y}{2} \right) \right] = x + c$
 (C) $\log \left[1 - \tan \left(\frac{x+y}{2} \right) \right] = x + c$ (D) None of these

Ans. B

Sol. -

Q19. The solution of $(dy/dx) = (4x + y + 1)^2$ is-

- (A) $4x - y + 1 = 2 \tan(2x - 2c)$ (B) $4x - y - 1 = 2 \tan(2x - 2c)$
 (C) $4x + y + 1 = 2 \tan(2x + 2c)$ (D) None of these

Ans. C

Sol. -

Q20. The solution of the differential equation $(\sin x + \cos x) dy + (\cos x - \sin x) dx = 0$ is-

- (A) $e^x (\sin x + \cos x) + c = 0$ (B) $e^y (\sin x + \cos x) = c$ (C) $e^y (\cos x - \sin x) = c$
 (D) $e^y (\sin x - \cos x) = c$

Ans. B

Sol. -

Q21. The solution of $(x\sqrt{1+y^2})dx + (y\sqrt{1+x^2})dy = 0$ is -

- (A) $\sqrt{1+x^2} + \sqrt{1+y^2} = c$ (B) $\sqrt{1+x^2} - \sqrt{1+y^2} = c$ (C) $(1+x^2)^{3/2} + (1+y^2)^{3/2} = c$
 (D) None of these

Ans. A

Sol. -

Q22. Solution of differential equation $x = 1 + xy \frac{dy}{dx} + \frac{(xy)^2}{2!} \left(\frac{dy}{dx} \right)^2 + \frac{(xy)^3}{3!} \left(\frac{dy}{dx} \right)^3 + \dots$ is-

- (A) $y = \log_e(x) + C$ (B) $y = (\log_e x)^2 + C$ (C) $y = \pm \sqrt{(\log_e x)^2 + 2C}$

(D) $xy = x^y + k$

Ans. C

Sol. -

Q23. The solution of the differential equation $xdy - ydx = \sqrt{x^2 + y^2} dx$ is

(A) $x + \sqrt{x^2 + y^2} = cx^2$ (B) $y - \sqrt{x^2 + y^2} = cx$ (C) $x - \sqrt{x^2 + y^2} = cx$

(D) $y + \sqrt{x^2 + y^2} = cx^2$

Ans. D

Sol. -

Q24. A curve C has the property that if the tangent drawn at any point P on C meets the coordinate axis at A and B, then P is the midpoint of AB. If the curve passes through the point (1,1) then the equation of the curve is-

(A) $xy = 2$ (B) $xy = 3$ (C) $xy = 1$ (D) None of these

Ans. C

Sol. -

Q25. A curve is such that the area of the region bounded by the co-ordinate axes, the curve & the ordinate of any point on it is equal to the cube of that ordinate. The curve represents

(A) a pair of straight lines (B) a circle (C) a parabola (D) an ellipse

Ans. C

Sol. -

Q26. In a chemical reaction a substance changes into another such that the rate of decomposition of a chemical substance x present at instant t is proportional to x itself i.e. amount of unchanged substance still present. If half of the substance present initially has been converted at the end of 1 minute then the time t in minutes at the end of which 99 % of the substance will have changed lies in the interval

(A) 5 and 6 (B) 6 and 7 (C) 7 and 8 (D) more than 10

Ans. B

Sol. -

Q27. The latus rectum of the conic passing through the origin and having the property that normal at each point (x, y) intersects the x - axis at $((x + 1), 0)$ is

(A) 1 (B) 2 (C) 4 (D) None of these

Ans. B

Sol. -

Q28. The equation of the curve through the point $(1, 0)$, whose slope is $\frac{y-1}{x^2+x}$, is-

- (A) $(y-1)(x+1)+2x=0$ (B) $2x(y-1)+x+1=0$ (C) $x(y-1)(x+1)+2=0$
(D) $x(y+1)+y(x+1)=0$

Ans. A

Sol. -

Q29. $\frac{dy}{dx} \left(\frac{1+\cos x}{y} \right) = -\sin x$ and $f\left(\frac{\pi}{2}\right) = -1$, then $f(0)$ is

- (A) 2 (B) 1 (C) 3 (D) 4

Ans. A

Sol. -

Q30. The solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$ is -

- (A) $y = x \ln x + x^2$ (B) $y = xe^{(x-1)}$ (C) $y = x \ln x + x$ (D) $y = \ln x + x$

Ans. C

Sol. -

Q31. The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with

- (A) variable radii and a fixed centre at $(0, 1)$
(B) variable radii and a fixed centre at $(0, -1)$
(C) fixed radius 1 and variable centres along the x-axis.
(D) fixed radius 1 and variable centres along the y-axis.

Ans. C

Sol. -

Q32. If $\frac{dy}{dx} = y + 3 > 0$ and $y(0) = 2$, then $y(\ln 2)$ is equal to :-

- (A) 13 (B) -2 (C) 7 (D) 5

Ans. C

Sol. -

Q33. Let I be the purchase value of an equipment and $V(t)$ be the value after it has been used for t years. The value $V(t)$ depreciates at a rate given by differential equation $\frac{dv(t)}{dt} = -k(T - t)$, where $k > 0$ is a constant and T is the total life in years of the equipment. Then value $V(T)$ of the equipment is :-

(A) $I - \frac{k(T-t)^2}{2}$ (B) e^{kT} (C) $T^2 - \frac{I}{k}$ (D) $I - \frac{kT^2}{2}$

Ans. D

Sol. -

Q34. The population $p(t)$ at time t of a certain mouse species satisfies differential equation $\frac{dp(t)}{dt} = 0.5p(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is :

(A) $\ln 18$ (B) $2 \ln 18$ (C) $\ln 9$ (D) $\frac{1}{2} \ln 18$

Ans. B

Sol. $\frac{dP(t)}{dt} = \frac{1}{2}P(t) - 450$

integrate

$$\int \frac{dP}{P - 900} = \int \frac{1}{2} dt$$

$$\ln |(P - 900)| = \frac{1}{2}t + C \dots \dots (1)$$

given $t = 0 \rightarrow P = 850$

$$\therefore C = \ln 50$$

from (1)

$$\ln |(P - 900)| = \frac{1}{2}t + \ln 50$$

$$\frac{1}{2}t = \ln \left| \left(\frac{P - 900}{50} \right) \right|$$

$$t = 2 \ln \left| \left(\frac{P - 900}{50} \right) \right|$$

at $P = 0$

$$t = 2 \ln \frac{900}{50}$$

$$t = 2 \ln 18$$

Q35. At present a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workesrs x is given by $\frac{dp}{dx} = 10 - 120\sqrt{x}$. If

the firm employs 25 more workers, then the new level of production of items is

- (A) 2500 (B) 3000 (C) 3500 (D) 4500

Ans. C

Sol. -

Q36. Let the population of rabbits surviving at a time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$. If $p(0) = 100$, then $p(t)$ equals:

- (A) $600 - 500 e^{t/2}$ (B) $400 - 300 e^{-t/2}$ (C) $400 - 300 e^{-t/2}$ (D) $300 - 200 e^{-t/2}$

Ans. C

Sol. -

Q37. If $(2 + \sin x) \frac{dy}{dx} + (y + 1) \cos x = 0$ and $y(0) = 1$, then $y\left(\frac{\pi}{2}\right)$ is equal to

- (A) $-\frac{1}{3}$ (B) $\frac{4}{3}$ (C) $\frac{1}{3}$ (D) $-\frac{2}{3}$

Ans. C

Sol. -

Q38. The solution of the equation $x \frac{dy}{dx} = y - x \tan\left(\frac{y}{x}\right)$ is -

- (A) $x \sin\left(\frac{x}{y}\right) + c = 0$ (B) $x \sin y + c = 0$ (C) $x \sin\left(\frac{y}{x}\right) = c$ (D) None of these

Ans. C

Sol. -

Q39. The equation of a curve passing through $(1, 0)$ for which the product of the abscissa of a point P & the intercept made by a normal at P on the x-axis equals twice the square of the radius vector of the point P, is

- (A) $x^2 + y^2 = x^4$ (B) $x^2 + y^2 = 2x^4$ (C) $x^2 - y^2 = 4x^4$ (D) $x^2 - y^2 = x^4$

Ans. A

Sol. -

Q40. The solution of the differential equation $(x^2 - y^2) dx + 2xy dy = 0$ is-

- (A) $x^2 + y^2 = cx$ (B) $x^2 - y^2 + cx = 0$ (C) $x^2 + 2xy = y^2 + cx$
(D) $x^2 + y^2 = 2xy + cx^2$

Ans. A

Sol. -

Q41. The x-intercept of the tangent to a curve is equal to the ordinate of the point of contact. The equation of the curve through the point (1, 1) is

- (A) $ye^{\frac{x}{y}} = e$ (B) $xe^{\frac{x}{y}} = e$ (C) $ye^{\frac{y}{x}} = e$ (D) $ye^{\frac{y}{x}} = e$

Ans. A

Sol. -

Q42. Which of the following equation is linear-

- (A) $\left(\frac{dy}{dx}\right)^2 + x^2\left(\frac{dy}{dx}\right)^2 = 0$ (B) $y = \frac{dy}{dx} + \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$ (C) $\frac{dy}{dx} + \frac{y}{x} = \log x$
(D) $y\frac{dy}{dx} - 4 = x$

Ans. C

Sol. -

Q43. The solution of the differential equation $\frac{dy}{dx} + y \cot x = 2 \cos x$ is-

- (A) $y \sin x + \cos 2x = 2c$ (B) $2y \sin x + \cos x = c$ (C) $y \sin x + \cos x = c$
(D) $2y \sin x + \cos 2x = c$

Ans. D

Sol. -

Q44. The solution of $(1 + y^2) dx = (\tan^{-1} y - x) dy$ is -

- (A) $xe^{\tan^{-1} y} = e^{\tan^{-1} y} (\tan^{-1} y - 1) + c$ (B) $xe^{\tan^{-1} y} = (\tan^{-1} y + 1) - c$
(C) $xe^{\tan^{-1} y} = (\tan^{-1} y - 1) + c$ (D) None of these

Ans. A

Sol. -

Q45. The solution of the equation $\frac{dy}{dx} + y \tan x = x^m \cos x$ is-

- (A) $(m+1)y = x^{m+1} \cos x + c (m+1) \cos x$ (B) $my = (x^m + c) \cos x$
(C) $y = (x^{m+1} + c) \cos x$ (D) None of these

Ans. A

Sol. -

Q46. The solution of the differential equation $\frac{dy}{dx} + \frac{3x^2}{1+x^3}y = \frac{\sin^2}{1+x^3}$ is -

- (A) $y(1+x^3) = x + \frac{1}{2} \sin 2x + c$ (B) $y(1+x^3) = cx + \frac{1}{2} \sin 2x$

$$(C) y(1 + x^3) = cx - \frac{1}{2} \sin 2x \quad (D) y(1 + x^3) = \frac{x}{2} - \frac{1}{4} \sin 2x + c$$

Ans. D

Sol. -

Q47. The solution of the differential equation $(1 + y^2) + \left(x - e^{\tan^{-1} y}\right) \frac{dy}{dx} = 0$, is

$$(A) xe^{2 \tan^{-1} y} = e^{\tan^{-1} y} + K \quad (B) (x - 2) = K e^{\tan^{-1} y} \quad (C) 2xe^{\tan^{-1} y} = e^{2 \tan^{-1} y} + K$$

$$(D) xe^{\tan^{-1} y} = \tan^{-1} y + K$$

Ans. C

Sol. -

Q48. A function $y = f(x)$ satisfies the condition $f'(x) \sin x + f(x) \cos x = 1$, $f(x)$ being bounded when $x \rightarrow 0$. If

$$I = \int_0^{\pi/2} f(x) dx \text{ then}$$

$$(A) \frac{\pi}{2} < I < \frac{\pi^2}{4} \quad (B) \frac{\pi}{4} < I < \frac{\pi^2}{2} \quad (C) 1 < I < \frac{\pi}{2} \quad (D) 0 < I < 1$$

Ans. A

Sol. -

Q49. The general solution of $\frac{dy}{dx} = 2y \tan x + \tan^2 x$ is

$$(A) y \cos^2 x = \frac{x}{2} - \frac{\sin 2x}{4} + C \quad (B) y \sec^2 x = \frac{x}{2} - \frac{\sin 2x}{4} + C$$

$$(C) y \cos^2 x = \frac{x}{2} - \frac{\cos 2x}{4} + C \quad (D) y \cos^2 x = \frac{x}{2} - \frac{\sin 2x}{2} + C$$

Ans. A

Sol. -

Q50. Let $f(x)$ be differentiable on the interval $(0, \infty)$ such that $f(1) = 1$ and

$$\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1 \text{ for each } \forall x > 0. \text{ Then } f(x) \text{ is}$$

$$(A) \frac{1}{3x} + \frac{2x^2}{3} \quad (B) \frac{-1}{3x} + \frac{4x^2}{3} \quad (C) \frac{-1}{x} + \frac{2}{x^2} \quad (D) \frac{1}{x}$$

Ans. A

Sol. -

Q51. For the differential equation $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$. If $y(1) = 1$, then x is given by :

(A) $1 - \frac{1}{y} + \frac{e^{\frac{1}{y}}}{e}$ (B) $4 - \frac{2}{y} - \frac{e^{\frac{1}{y}}}{e}$ (C) $3 - \frac{1}{y} + \frac{e^{\frac{1}{y}}}{e}$ (D) $1 + \frac{1}{y} - \frac{e^{\frac{1}{y}}}{e}$

Ans. D

Sol. -

Q52. Let $y(x)$ be the solution of the differential equation $(x \log x) \frac{dy}{dx} + y = 2x \log x, (x \geq 1)$. Then $y(e)$ is equal to :

(A) e (B) 0 (C) 2 (D) $2e$

Ans. C

Sol. $\ln x \frac{dy}{dx} + \frac{y}{x} = 2 \ln x$

$$\int d(y \ln x) = 2 \int \ln x dx$$

$$y \ln x = 2x(\ln x - 1) + c$$

$$y(e) = 2 \quad \text{as } c = 2$$

Q53. Let $y = y(x)$ be the solution of the differential equation $\sin x \frac{dy}{dx} + y \cos x = 4x, x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to

(A) $-\frac{8}{9}\pi^2$ (B) $-\frac{4}{9}\pi^2$ (C) $\frac{4}{9\sqrt{3}}\pi^2$ (D) $\frac{-8}{9\sqrt{3}}\pi^2$

Ans. A

Sol. -

Q54. If the differentiable equation $\frac{dy}{dx} - y = y^2(\sin x + \cos x)$ with $y(0) = 1$ then $y(\pi)$ has the value equal to

(A) e^π (B) $-e^\pi$ (C) $e^{-\pi}$ (D) $-e^{-\pi}$

Ans. A

Sol. -

Q55. Solution of the differential equation $\cos x dy = y(\sin x - y)dx, 0 < x < \frac{\pi}{2}$ is :

(A) $\sec x = (\tan x + c)y$ (B) $y \sec x = \tan x + c$ (C) $y \tan x = \sec x + c$

(D) $\tan x = (\sec x + c)y$

Ans. A

Sol. -

Q56. If a curve $y = f(x)$ passes through the point $(1, -1)$ and satisfies the differential equation, $y(1 + xy)dx = x dy$, then $f\left(-\frac{1}{2}\right)$ is equal to :

- (A) $2/5$ (B) $4/5$ (C) $-\frac{2}{5}$ (D) $-\frac{4}{5}$

Ans. B

Sol. -

Q57. The solution of the differential equation, $e^x(x + 1)dx + (ye^y - xe^x)dy = 0$ with initial condition $f(0) = 0$, is

- (A) $xe^x + 2y^2e^y = 0$ (B) $2xe^x + y^2e^y = 0$ (C) $xe^x - 2y^2e^y = 0$ (D) $2xe^x - y^2e^y = 0$

Ans. B

Sol. -

Q58. The solution of the differential equation, $x^2 \frac{dy}{dx} \cdot \cos \frac{1}{x} - y \sin \frac{1}{x} = -1$, where $y \rightarrow -1$ as $x \rightarrow \infty$ is

- (A) $y = \sin \frac{1}{x} - \cos \frac{1}{x}$ (B) $y = \frac{x+1}{x \sin \frac{1}{x}}$ (C) $y = \cos \frac{1}{x} + \sin \frac{1}{x}$ (D) $y = \frac{x+1}{x \cos \frac{1}{x}}$

Ans. A

Sol. -

Q59. If $x \frac{dy}{dx} = y (\log y - \log x + 1)$, then the solution of the equation is -

- (A) $y \log \left(\frac{x}{y}\right) = cx$ (B) $x \log \left(\frac{y}{x}\right) = cy$ (C) $\log \left(\frac{y}{x}\right) = cx$ (D) $\log \left(\frac{x}{y}\right) = cy$

Ans. C

Sol. -

Q60. The solution of the differential equation $ydx + (x + x^2 y) dy = 0$ is-

- (A) $-\frac{1}{xy} = C$ (B) $-\frac{1}{xy} + \log y = C$ (C) $\frac{1}{xy} + \log y = C$ (D) $\log y = Cx$

Ans. B

Sol. -

Exercise J02

Q1. Order, degree and formation of a differentail equation

(a) Form the differential equation of the family of curves represented by, $c(y + c)^2 = x^3$; where c is any arbitrary constant.

(b) Form a differential equation for the family of curves represented by $ax^2 + by^2 = 1$, where a & b are arbitrary constants.

(c) Obtain the differential equation of the family of circles $x^2 + y^2 + 2gx + 2fy + c = 0$; where g , f & c are arbitrary constants.

(d) Obtain the differential equation associated with the primitive, $y = c_1e^{3x} + c_2e^{2x} + c_3e^x$, where c_1, c_2, c_3 are arbitrary constants.

Ans. (a) $12y(y')^2 = x[8(y')^3 - 27]$; (b) $xy \frac{d^2x}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} = 0$;
 (c) $[1 + (y')^2] \cdot y''' - 3y'(y'')^2 = 0$ (d) $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$

Sol. -

Q2. Solve the following differential equation for

$$\frac{\ln(\sec x + \tan x)}{\cos x} dx = \frac{\ln(\sec y + \tan Y)}{\cos Y} dy$$

Ans. $\ln^2(\sec x + \tan x) - \ln^2(\sec y + \tan y) = c$

Sol. -

Q3. Solve the following differential equation for

$$(1 - x^2)(1 - y) dx = xy(1 + y) dy$$

Ans. $\ln x(1 - y)^2 = c - \frac{1}{2}y^2 - 2y + \frac{1}{2}x^2$

Sol. -

Q4. Solve the following differential equation

$$\frac{dy}{dx} + \frac{\sqrt{(x^2 - 1)(y^2 - 1)}}{xy} = 0$$

Ans. $\sqrt{x^2 - 1} - \sec^{-1} x + \sqrt{y^2 - 1} = c$

Sol. -

Q5. Solve the following differential equation

$$\frac{dy}{dx} = \sin(x + y) + \cos(x + y)$$

Ans. $\ln \left[1 + \tan \frac{x + y}{2} \right] = x + c$

Sol. -

Q6. Solve the following differential equation

$$\frac{dy}{dx} = \frac{x(2 \ln x + 1)}{\sin y + y \cos y}$$

Ans. $y \sin y = x^2 \ln x + c$

Sol. -

Q7. Solve the following differential equation

$$(a) \frac{dy}{dx} + \sin \frac{x+y}{2} = \sin \frac{x-y}{2} \quad (b) \sin x \cdot \frac{dy}{dx} = y \cdot \ln y \text{ if } y = e, \text{ when } x = \frac{\pi}{2}$$

Ans. (a) $\ln \left| \tan \frac{y}{4} \right| = c - 2 \sin \frac{x}{2}$, (b) $y = e^{\tan(x/2)}$

Sol. -

Q8. The population P of a town decreases at a rate proportional to the number by which the population exceeds 1000, proportionality constant being $k > 0$. Find

(a) Population at any time t, given initial population of the town being 2500.

(b) If 10 years later the population has fallen to 1900, find the time when the population will be 1500.

(c) Predict about the population of the town in the long run.

Ans. (a) $P = 1000 + 1500e^{-kt}$ where $k = \frac{1}{10} \ln \left(\frac{5}{3} \right)$; (b) $T = 10 \log_{5/3}(3)$; (c) $P = 1000$ as $t \rightarrow \infty$

Sol. -

Q9. It is known that the decay rate of radium is directly proportional to its quantity at each given instant. Find the law of variation of a mass of radium as a function of time if at $t = 0$, the mass was m_0 and during time t_0 α % of the original mass of radium decay.

Ans. $m = m_0 e^{-kt}$ where $k = -\frac{1}{t_0} \ln \left(1 - \frac{\alpha}{100} \right)$

Sol. $\frac{dm}{dt} = -\lambda m$

$$\frac{1}{m} dm = -\lambda dt$$

$$\ln m = -\lambda t + c$$

$$m = ke^{-\lambda t} \quad (\text{at } t = 0, m = m_0)$$

$$\Rightarrow k = m_0$$

$$m = m_0 e^{-\lambda t} \quad \left(\text{at } t = t_0, m = m_0 - \frac{\alpha m_0}{100} \right)$$

$$\Rightarrow \lambda = \frac{-1}{t_0} \ln \left(1 - \frac{\alpha}{100} \right)$$

Q10. A normal is drawn at a point $P(x, y)$ of a curve. It meets the x -axis at Q . If PQ is of constant length k , then show that the differential equation describing such curves is, $y \frac{dy}{dx} = \pm \sqrt{k^2 - y^2}$. Find the equation of such a curve passing through $(0, k)$.

Ans. $x^2 + y^2 = k^2$

Sol. -

Q11. $\frac{x dx - y dy}{x dy - y dx} \sqrt{\frac{1 + x^2 - y^2}{x^2 - y^2}}$

Ans. $\sqrt{x^2 - y^2} + \sqrt{1 + x^2 - y^2} = \frac{c(x + y)}{\sqrt{x^2 - y^2}}$

Sol. -

Q12. A curve is such that the length of the polar radius of any point on the curve is equal to the length of the tangent drawn at this point. Form the differential equation and solve it to find the equation of the curve.

Ans. $y = kx$ or $xy = c$

Sol. -

Q13. Tangent is drawn at the point (x_i, y_i) on the curve $y = f(x)$, which intersects the x -axis at $(x_{i+1}, 0)$. Now again a tangent is drawn at (x_{i+1}, y_{i+1}) on the curve which intersects the x -axis at $(x_{i+2}, 0)$ and the process is repeated n times i.e. $i = 1, 2, 3, \dots, n$.

(a) If $x_1, x_2, x_3, \dots, x_n$ form an arithmetic progression with common difference equal to $\log_2 e$ and curve passes through $(0, 2)$, then find the equation of the curve.

(b) If $x_1, x_2, x_3, \dots, x_n$ form a geometric progression with common ratio equal to 2 and the curve passes through $(1, 2)$, then find the equation of the curve.

Ans. (a) $y = 2^{1-x}$; (b) $xy = 2$

Sol. -

Q14. Given $y(0) = 2000$ and $\frac{dy}{dx} = 32000 - 20y^2$, then find the value of $\lim_{x \rightarrow \infty} y(x)$.

Ans. 40

Sol. -

Q15. Let $f(x)$ is a continuous function which takes positive values for $x \geq 0$ and satisfy

$$\int_0^x f(t) dt = x \sqrt{f(x)} \text{ with}$$

$f(1) = \frac{1}{2}$. Find the value of $f(\sqrt{2} + 1)$.

Ans. $1/4$

Sol. -

Q16. $\frac{dy}{dx} = \frac{x^2 + xy}{x^2 + y^2}$

Ans. $c(x - y)^{2/3} (x^2 + xy + y^2)^{1/6} = \exp \left[\frac{1}{\sqrt{3}} \tan^{-1} \frac{x + 2y}{x\sqrt{3}} \right]$ where $\exp x \equiv e^x$

Sol. -

Q17. Find the equation of a curve such that the projection of its ordinate upon the normal is equal to its abscissa.

Ans. $\frac{y^2 \pm y\sqrt{y^2 - x^2}}{x^2} = \ln \left| \left(y \pm \sqrt{y^2 - x^2} \right) \cdot \frac{c^2}{x^3} \right|$, where same sign has to be taken .

Sol. -

Q18. The light rays emanating from a point source situated at origin when reflected from the mirror of a search light are reflected as beam parallel to the x-axis. Show that the surface is parabolic, by first forming the differential equation and then solving it.

Ans. -

Sol. -

Q19. The perpendicular from the origin to the tangent at any point on a curve is equal to the abscissa of the point of contact. Find the equation of the curve satisfying the above condition and which passes through (1, 1).

Ans. $x^2 + y^2 - 2x = 0$

Sol. -

Q20. Show that the equation of the curve intersecting with the x- axis at the point $x = 1$ and for which the length of the subnormal at any point of the curve is equal to the arithmetic mean of the co-ordinates of this point is

$$(y - x)^2(x + 2y) = 1.$$

Ans. -

Sol. -

Q21. $\left[x \cos \frac{y}{x} + y \sin \frac{y}{x} \right] y = \left[y \sin \frac{y}{x} - x \cos \frac{y}{x} \right] x \frac{dy}{dx}$

Ans. $xy \cos \frac{y}{x} = c$

Sol. -

Q22. Find the curve for which any tangent intersects the y-axis at the point equidistant from the point of tangency and the origin.

Ans. $x^2 + y^2 = cx$

Sol. -

Q23. $(x - y) dy = (x + y + 1) dx$

Ans. $\arctan \frac{2y+1}{2x+1} = \ln c \sqrt{x^2 + y^2 + x + y + \frac{1}{2}}$

Sol. -

Q24. $\frac{dy}{dx} = \frac{x + y + 1}{2x + 2y + 3}$

Ans. $x + y + \frac{4}{3} = ce^{3(x-2y)}$

Sol. -

Q25. Show that the curve such that the distance between the origin and the tangent at an arbitrary point is equal to the distance between the origin and the normal at the same point, $\sqrt{x^2 + y^2} = ce^{\pm \tan^{-1} \frac{y}{x}}$

Ans. -

Sol. -

Q26. Let $y = y(t)$ be a solution to the differential equation $y' + 2ty = t^2$, then find $\lim_{t \rightarrow \infty} \frac{y}{t}$.

Ans. $\frac{1}{2}$

Sol. -

Q27. $\frac{dy}{dx} + \frac{x}{1+x^2}y = \frac{1}{2x(1+x^2)}$

Ans. $y\sqrt{1+x^2} = c + \frac{1}{2} \ln \left[\tan \frac{1}{2} \arctan x \right]$ Another form is $y\sqrt{1+x^2} = c + \frac{1}{2} \ln \frac{\sqrt{1+x^2} - 1}{x}$

Sol. -

Q28. $(1 - x^2) \frac{dy}{dx} + 2xy = x(1 - x^2)^{1/2}$

Ans. $y = c(1 - x^2) + \sqrt{1 - x^2}$

Sol. -

Q29. Find the curve such that the area of the trapezium formed by the co-ordinate axes, ordinate of an arbitrary point & the tangent at this point equals half the square of its abscissa.

Ans. $y = cx^2 \pm x$

Sol. -

Q30. A curve in the first quadrant is such that the area of the triangle formed in the first quadrant by the x-axis, a tangent to the curve at any of its point P and radius vector of the point P is 2 sq. units. If the curve passes through (2, 1), find the equation of the curve.

Ans. $xy = 2$

Sol. -

Q31. Find the curve possessing the property that the intercept, the tangent at any point of a curve cuts off on the y-axis is equal to the square of the abscissa of the point of tangency.

Ans. $y = cx - x^2$

Sol. -

Q32. $x \frac{dy}{dx} - y = 2x^2 \operatorname{cosec} 2x$

Ans. $y = cx + x \ln \tan x$

Sol. -

Q33. $(1 + y^2) dx = (\tan^{-1} y - x) dy$

Ans. $x = ce^{-\arctan y} + \arctan y - 1$

Sol. -

Q34. Find the differentiable function which satisfies the equation $f(x) = -$

$$\int_0^x f(t) \tan t dt + \int_0^x \tan(t-x) dt$$

where $x \in (-\pi/2, \pi/2)$

Ans. $\cos x - 1$

Sol. -

Q35. Find all functions $f(x)$ defined on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with real values and has a primitive $F(x)$ such that $f(x) + \cos x \cdot F(x) = \frac{\sin 2x}{(1 + \sin x)^2}$. Find $f(x)$.

Ans. $f(x) = -\frac{2 \cos x}{(1 + \sin x)^2} - Ce^{\sin x} \cdot \cos x$

Sol. -

Q36. A tank contains 100 litres of fresh water. A solution containing 1 gm/litre of soluble lawn fertilizer runs into the tank at the rate of 1 lit/min, and the mixture is pumped out of the tank at the rate of 3 litres/min. Find the time when the amount of fertilizer in the tank is maximum.

Ans. $24\frac{7}{9}$

Sol. -

Q37. A tank with a capacity of 1000 litres originally contains 100 gms of salt dissolved in 400 litres of water. Beginning at time $t = 0$ and ending at time $t = 100$ minutes, water containing 1 gm of salt per litre enters the tank at the rate of 4 litres per minute, and the well mixed solution is drained from the tank at a rate of 2 litre/minute. Find the differential equation for the amount of salt y in the tank at time t .

Ans. $\frac{dy}{dt} = 4 - \frac{y}{200 + t}$

Sol. -

Q38. A curve passes through $(2, 0)$ and the slope of tangent at point $P(x, y)$ equals $\frac{(x+1)^2 + y - 3}{(x+1)}$. Find the equation of the curve and area enclosed by the curve and the x -axis in the fourth quadrant.

Ans. $y = c(1 - x^2) + \sqrt{1 - x^2}$

Sol. -

Q39. If length of tangent at any point on the curve $y = f(x)$ intercepted between the point and the x -axis is of length 1. Find the equation of the curve.

Ans. $\sqrt{1 - y^2} + \ln \left| \frac{1 - \sqrt{1 - y^2}}{y} \right| = \pm x + c$

Sol. -

Q40. (a) Let f be a real valued differentiable function on $\mathbb{R}$ (the set of all real numbers) such that $f(1) = 1$. If the y -intercept of the tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of P , then the value of $f(-3)$ is equal to

(b) Let $y'(x) + y(x)g'(x) = g(x)g'(x)$, $y(0) = 0$, $x \in \mathbb{R}$, where $f'(x)$ denotes $\frac{df(x)}{dx}$ and $g(x)$ is a given

nonconstant differentiable function on $\mathbb{R}$ with $g(0) = g(2) = 0$. Then the value of $y(2)$ is

Ans. (a) 9, (b) 0

Sol. -

Q41. $2\frac{dy}{dx} - y \sec x = y^3 \tan x$

Ans. $\frac{1}{y^2} = -1 + (c + x) \cot \left(\frac{x}{2} + \frac{\pi}{4} \right)$

Sol. -

Q42. $y(2xy + e^x) dx - e^x dy = 0$

Ans. $y^{-1} e^x = c - x^2$

Sol. -

Q43. $(x - y^2) dx + 2xy dy = 0$

Ans. $y^2 + x \ln ax = 0$

Sol. -

Q44. $x \frac{dy}{dx} + y \ln y = xye^x$

Ans. $x \ln y = e^x (x - 1) + c$

Sol. -

Q45. $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$

Ans. $\sin y = (e^x + c)(1+x)$

Sol. -

Q46. $\frac{dy}{dx} = \frac{e^y}{x^2} - \frac{1}{x}$

Ans. $cx^2 + 2xe^{-y} = 1$

Sol. -

Q47. $\frac{dy}{dx} = e^{x-y} (e^x - e^y)$

Ans. $e^y = c \cdot \exp(-e^x) + e^x - 1$

Sol. -

Q48. Identify the statement(s) which is/are True.

(A) $f(x, y) = e^{y/x} + \tan \frac{y}{x}$ is a homogeneous function of degree zero

(B) $x \cdot \ln \frac{y}{x} dx + \frac{y^2}{x} \sin^{-1} \frac{y}{x} dy = 0$ is a homogeneous differential equation of degree one

(C) $f(x, y) = x^2 + \sin x \cdot \cos y$ is not homogeneous function.

(D) $(x^2 + y^2) dx - (xy^2 - y^3) dy = 0$ is a homogeneous differential equation.

Ans. A, B, C

Sol. -

Q49. A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx} \cdot \sin x - y \cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \rightarrow 0$ as $x \rightarrow \infty$ then the statement which is correct is

- (A) $\lim_{x \rightarrow 0} f(x) = 1$ (B) $\int_0^{\pi/2} f(x) dx$ is less than $\frac{\pi}{2}$ (C) $\int_0^{\pi/2} f(x) dx$ is greater than unity
- (D) $f(x)$ is an odd function

Ans. A, B, C

Sol. -

Q50. The tangent at any point P on a curve $f(x,y) = 0$ cuts the y-axis at T. If the distance of point T from P equals the distance of T from the origin then the curve with this property represents a family of circles. Which of the following are correct?

- (A) any arbitrary line $y = mx$ cuts every member of this family at the points where the slopes of these members are equal
- (B) $f(x,y) = 0$ is orthogonal to the family of circles $x^2 + y^2 - ky = 0 \forall x \in R$
- (C) If $f(x,y) = 0$ passes through (2, 2) then the intercept made by its director circle on the y-axis is equal to 8
- (D) If $f(x,y) = 0$ passes through (-1, 1), then image of its centre in the line $y = x$ is (1, 0)

Ans. A, B

Sol. Equation of tangent $(y - y) = m(X - x)$

Put $X = 0$, $Y = y - mx$

Now, $(OT)^2 = PT^2$

$$\Rightarrow (y - mx)^2 = x^2 + m^2 x^2$$

$$\Rightarrow y^2 + m^2 x^2 - 2mxy = x^2 + m^2 x^2$$

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

On solving $y^2 + x^2 = cx$

If it passes through (2, 2) $\Rightarrow c = 4$

$$(x - 2)^2 + y^2 = 4$$

Its director circle will be

$$(x - 2)^2 + y^2 = 8$$

Put $x = 0$, $y^2 = r \Rightarrow y = 2, -2$

Intercept on y-axis is 4

Q51. Let C be the family of curves $f(x, y, c) = 0$ (no member of C is x-axis) such that length of subnormal at any point P(x, y) on the curve C is equal to four times that of the length of subtangent at the same point. Which of the following statement(s) is(are) correct ?

- (A) Equation of the line with positive y-intercept passing through (4, 2) and perpendicular to the curve C is $x + 2y = 8$.
- (B) Orthogonal trajectory of C is family of parallel lines having gradient ± 2 .
- (C) Order and degree of the differential equation of family of curves C are 1 and 2 respectively.
- (D) Differential equation of family of curves is $2y' \pm x = 0$.

Ans. A, C

Sol. -

Q52. Let C be a curve such that the tangent at any point P on it meets x-axis and y-axis at A and B respectively. If $BP : PA = 3:1$ and the curve passes through the point (1, 1), then

- (A) The curve passes through $\left(2, \frac{1}{8}\right)$
- (B) Equation of normal to the curve at (1, 1) is $3y - x = 2$
- (C) The differential equation for the curve is $3y' + xy = 0$
- (D) The differential equation for the curve is $xy' + 3y = 0$

Ans. D, B, A

Sol. -

Paragraph

A curve $y = f(x)$ satisfies the differential equation $(1 + x^2)\frac{dy}{dx} + 2yx = 4x^2$ and passes through the origin.

\n

Q53. For the function $y = f(x)$ which one of the following does not hold good?

- (A) $f(x)$ is a rational function (B) $f(x)$ has the same domain and same range.
- (C) $f(x)$ is a transcendental function (D) $y = f(x)$ is a bijective mapping.

Ans. C

Sol. -

Q54. The area enclosed by $y = f^{-1}(x)$, the x-axis and the ordinate at $x = 2/3$ is

- (A) $2 \ln 2$ (B) $\frac{4}{3} \ln 2$ (C) $\frac{3}{2} \ln 2$ (D) $\frac{1}{3} \ln 2$

Ans. C

Sol. -

Q55. The function $y = f(x)$

- (A) is strictly increasing $\forall x \in \mathbb{R}$. (B) is such that it has a minima but no maxima.
(C) is such that it has a maxima but no minima. (D) has no inflection point.

Ans. A

Sol. -

Paragraph

Let C be a curve $y = f(x)$ passing through $M(-\sqrt{3}, 1)$ such that the y -intercept of the normal at any point $P(x, y)$ on the curve C is equal to the distance of P from the origin.

Q56. The area enclosed by the curve C and the co-ordinate axes is equal to

- (A) $\frac{1}{6}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

Ans. D

Sol. -

Q57. Which one of the following statement is correct?

- (A) $f(x)$ has exactly two points of inflection. (B) $f(x)$ has maxima but no minima.
(C) The equation $f(x) = -2$ has two distinct real solutions.
(D) Inclination of the tangent line drawn to $f(x)$ at $(1, 0)$ is $\tan^{-1}1$.

Ans. D

Sol. -

Q58. The shortest distance between the curve C and the circle $x^2 + y^2 - 2y = 0$ is equal to

- (A) $\frac{1}{2}$ (B) $\sqrt{2} - 1$ (C) $\sqrt{6} - 1$ (D) $\sqrt{5} - 1$

Ans. B

Sol. -

Q59. Match the column type

Column-I	Column-II
(A) Solution of differential equation $[3x^2y + 2xy - e^x(1+x)] dx$	(P) $y^2(x^2 + 1 + ce^{x^2}) = 1$

	$+ (x^3 + x^2)dy = 0$ is	
Solution of differential (B) equation $ydx - xdy - 3xy^2 e^{x^2} dx = 0$ is	(Q) $(x^2 + x^3)y - xe^x = c$	
Solution of differential (C) equation $\frac{dy}{dx} = xy(x^2y^2 - 1)$ is	(R) $\frac{x}{y} - \frac{3}{2}e^{x^2} = c$	
Solution of differential (D) equation $\frac{dy}{dx} (x^2y^3 + xy) = 1$ is	(S) $\frac{1}{x} = 2 - y^2 + ce^{-y^2/2}$	
	(T) $\frac{2}{x} = 1 - y^2 + ce^{-y^2/2}$	

(A) A – Q, B – R, C – P, D – S (B) A – S, B – R, C – P, D – Q

(C) A – Q, B – P, C – R, D – S (D) A – Q, B – R, C – S, D – P

Ans. A

Sol. (A) $[3x^2y + 2xy - e^x(1+x)] dx + (x^3 + x^2)dy = 0$

$$y(3x^2 + 2x) + (x^3 + x^2)dy = e^x(1+x)dx$$

$$\int d[y(x^3 + x^2)] = \int (x+1)e^x dx$$

$$y(x^3 + x^2) = (x+1)e^x - e^x + c$$

$$y(x^3 + x^2) - xe^x = c$$

$$(B) ydx - xdy - 3xy^2 e^{x^2} dx = 0$$

$$ydx - xdy = 3xy^2 e^{x^2} dx$$

$$\int \frac{ydx - xdy}{y^2} = \int 3xe^{x^2} dx$$

$$x^2 = t$$

$$2xdx = dt$$

$$\frac{x}{y} = \frac{3}{2} \int e^t dt$$

$$\frac{x}{y} = \frac{3}{2} e^{x^2} + C$$

$$(C) \frac{dy}{dx} = xy(x^2y^2 - 1)$$

$$\frac{dy}{dx} = x^3y^3 - xy$$

$$\frac{dy}{dx} + xy = x^3y^3$$

$$\frac{1}{y^2} = t \quad - 2y^{-2-1} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{-1}{2} \frac{dt}{dx} + tx = x^3$$

$$I.F. = e^{\int -2x dx} = e^{-x^2}$$

$$\int e^{-x^2} \frac{dt}{dx} - 2xe^{-x^2}t = \int -2x^3e^{-x^2}$$

$$-x^2 = P$$

$$\Rightarrow -2x dx = dp$$

$$te^{-x^2} = - \int Pe^p dp$$

$$\frac{e^{-x^2}}{y^2} = -e^P (P - 1) + C$$

$$\frac{e^{-x^2}}{y^2} = -e^{-x^2} (-x^2 - 1) + C$$

$$\frac{1}{y^2} = x^2 + 1 + ce^{x^2}$$

$$(D) \frac{dy}{dx} (x^2y^3 + xy) = 1$$

$$\frac{dx}{dy} = x^2y^3 + xy$$

$$\frac{dx}{dy} - xy = x^2y^3$$

$$\frac{1}{x^2} \frac{dx}{dy} - \frac{y}{x} = y^3 \quad \frac{-1}{x} = t$$

$$\frac{1}{x^2} \frac{dx}{dy} = \frac{dt}{dy}$$

$$\frac{dt}{dy} + ty = y^3$$

$$I.F. = e^{\int y dy} = e^{\frac{y^2}{2}}$$

$$\int e^{\frac{y^2}{2}} \frac{dy}{dy} + tye^{\frac{y^2}{2}} = \int y^3 e^{\frac{y^2}{2}}$$

$$te^{\frac{y^2}{2}} = \int 2Pe^P dP + C$$

$$te^{\frac{y^2}{2}} = 2P^P (P - 1) + C$$

$$\frac{-e^{\frac{y^2}{2}}}{x} = 2e^{\frac{y^2}{2}} \left(\frac{y^2}{2} - 1 \right) + C$$

$$\frac{-1}{x} = 2 \left(\frac{y^2 - 2}{2} \right) + ce^{\frac{-y^2}{2}}$$

$$\frac{1}{x} = 2 - y^2 + ce^{-\frac{y^2}{2}}$$